AutoGuardian Car Anti-Theft System: Hardware Components List

Component	Description / Purpose
ESP32 Dev Board (with Wi-Fi & Bluetooth)	Main microcontroller that manages logic, GSM/GPS communication, and sensor inp
SIM800L GSM Module	Used for sending/receiving SMS or HTTP data via cellular networks.
GPS Module (e.g., NEO-6M)	Provides real-time location tracking of the vehicle.
5V Relay Module (Automotive Rated 30A)	Acts as a switch to control the fuel pump relay for kill switch functionality.
Flyback Diode (1N4007 or similar)	Protects ESP32 from voltage spikes caused by the relay coil when switching.
Voltage Regulator (e.g., AMS1117 12V to 5V)	Steps down car battery voltage to a safe 5V level for the ESP32.
Buzzer (Passive)	Provides audible alerts for arming/disarming or detected events.
PIR Motion Sensor (HC-SR501)	Detects human motion near the car for intrusion detection.
TTP223 Touch Sensor	Acts as tamper/touch detector on the system housing.
OLED Display (0.96" I2C)	Displays system status (armed, disarmed, GPS lock, etc.).
2-Channel Relay Module	Used to control additional features (e.g., ignition line or auxiliary).
Resistors, Diodes, Capacitors	Basic passive components for interfacing and protection.
Dupont Jumper Wires	Used for connections during prototyping and testing stages.
PCB or Perfboard	Mounts and connects components in a semi-permanent fashion for real-world deplo
Project Enclosure Box	Protects electronics from dust, heat, tampering in vehicle installation.
Fuse Holder + Fuse (e.g., 5A)	Prevents overcurrent damage to the circuit from vehicle power.
SIM Card (1.8V/3V)	Required for GSM communication on SIM800L.
GPS Antenna (Active)	Enhances GPS signal reception when placed inside the car dashboard.